RED SEAL EXAM PREP

313A Refrigeration & AC Mechanic

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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Mock Exam — answer all questions, then check the key

Q1. **A low-pressure control (LPC) is set to cut out at 20 PSI and cut in at 45 PSI on an R-22 system. The LPC trips repeatedly. The MOST likely cause is:**

A) Condenser fan failure causing high head pressure

B) Compressor valves leaking

C) Thermostat setting too cold

D) Low charge causing suction pressure to fall

Q2. **A refrigeration contractor is servicing the rack system in a supermarket owned by a private grocery chain, on privately owned land in a Canadian city. Which halocarbon rules govern how the refrigerant is handled on that job?**

A) The Transportation of Dangerous Goods Regulations and the TC recovery cylinder specification

B) The Federal Halocarbon Regulations, 2022, made under the Canadian Environmental Protection Act, 1999

C) The federal Ozone-depleting Substances and Halocarbon Alternatives Regulations

D) The halocarbon regulation of the province or territory where the work is done

Q3. **Why can a small mass of refrigerant absorb so much heat in the evaporator?**

A) Because the compressor pushes it through the coil faster than a liquid

B) Because a change of state absorbs far more heat than a temperature rise

C) Because the pressure drop at the metering device chills it below the air

D) Because its temperature rises steadily from the coil inlet to the outlet

Q4. **A single-phase rooftop condensing unit's compressor hums and trips its overload on every start attempt. During a start attempt, voltage measured at the outdoor disconnect holds at 240 V while voltage measured at the compressor terminals collapses to about 190 V. What does the difference between those two readings localize?**

A) A supply problem at the utility transformer feeding the building

B) A failed start capacitor unable to give the motor its starting torque

C) High resistance between the disconnect and the compressor terminals

D) A seized compressor drawing locked rotor current until it overheats

Q5. **Refrigerants R-717 (ammonia) and R-744 (carbon dioxide) both carry numbers in the 700 series. What does that series indicate?**

A) A hydrocarbon refrigerant, numbered by its number of carbons

B) A zeotropic blend, numbered in the order it was registered

C) An inorganic compound, numbered as 700 plus its molecular mass

D) A high-pressure refrigerant, numbered by its working pressure

Q6. **What is the difference between refrigerant recovery, reclaim, and recycling?**

A) Recovery, reclaim, and recycling are all the same process with different names used by different manufacturers

B) Recovery: venting refrigerant safely through a filter. Reclaim: any reuse of recovered refrigerant. Recycling: destroying old refrigerant

C) Recovery: removing to a container. Recycling: basic cleanup for reuse. Reclaiming: reprocessing to virgin purity

D) Recovery and reclaim are identical; recycling means mixing refrigerants for reuse

Q7. **What is the purpose of a phase monitor relay in a three-phase refrigeration system?**

A) To monitor refrigerant phase changes and shut down the system if the refrigerant is undercharged

B) To balance the electrical load across three phases to reduce energy consumption

C) To protect three-phase motors from phase loss, phase reversal, and phase imbalance

D) To convert single-phase power to three-phase for use with commercial compressors

Q8. **After repairing a leak on a commercial condensing unit, a technician pressure tests with dry nitrogen at the test pressure marked on the nameplate, holds the test overnight without loss, releases the nitrogen to atmosphere, closes the system and weighs in the nameplate charge. On start-up the unit cools poorly and repeatedly trips its high-pressure control. What is the most likely cause?**

A) Nitrogen left in at atmospheric pressure because the system was not evacuated

B) Compressor valves damaged by holding the nitrogen at test pressure overnight

C) Moisture that entered during the repair and is freezing at the metering device

D) A filter drier that was not replaced and has plugged with debris from the repair

Q9. **A hermetic compressor will not run. Winding-to-winding resistance readings are normal and balanced, but an insulation resistance test reads very low from a winding to the compressor shell. What does this indicate?**

A) A winding is grounded to the shell, so the compressor is replaced

B) Normal for a hermetic motor, since the windings return through the shell

C) The run capacitor has failed and is grounding the start winding

D) The internal overload is open and will reset once the shell cools

Q10. **A split system air conditioner is not cooling and the outdoor unit is not running, but the indoor air handler fan is running. What should be checked first?**

A) Refrigerant charge level — low charge prevents outdoor unit from starting

B) Power to the outdoor unit: breaker, disconnect, and contactor

C) The evaporator coil for ice formation blocking airflow

D) The TXV sensing bulb position

Q11. **Which component in a refrigeration system is responsible for the pressure drop that causes the refrigerant to change from high-pressure liquid to low-pressure liquid/vapor mixture?**

A) The filter drier

B) The metering device

C) The compressor

D) The evaporator

Q12. **R-410A (GWP 2088) is the focus of regulatory efforts to cut refrigerant emissions. Which refrigerant is designed to take its place in new residential air conditioning equipment?**

A) R-454B — an A2L refrigerant made from R-32 and the hydrofluoroolefin R-1234yf

B) R-22 — an HCFC refrigerant brought back for new equipment because its GWP is lower

C) R-125 — a single-component HFC with a lower GWP than the R-410A blend

D) CO_2 (R-744) — a non-flammable low-GWP refrigerant with pressures close to R-410A's

Q13. **What is the function of a high-pressure cutout switch in a refrigeration/air conditioning system?**

A) To limit compressor speed when head pressure is elevated

B) To bypass the TXV when discharge pressure is too high

C) To open the relief valve on the refrigerant cylinder when pressure exceeds maximum

D) To shut down the compressor when discharge pressure exceeds a safe limit

Q14. **What type of oil is required for systems using HFC refrigerants (such as R-134a, R-404A, R-410A)?**

A) POE (Polyolester) oil — HFCs are not miscible with mineral oil

B) Alkylbenzene oil — this is the universal oil compatible with all refrigerant types

C) PAG (Polyalkylene Glycol) oil — this is standard for all refrigeration applications

D) Mineral oil — the same oil used with R-12 and R-22 systems

Q15. **What is the purpose of a check valve in a refrigeration system, and where is it typically installed?**

A) To prevent oil from leaving the compressor — installed at the compressor suction port

B) To regulate refrigerant flow rate — installed in the liquid line as a primary metering device

C) To prevent backflow during the off-cycle — installed in heat pump discharge lines

D) To limit maximum system pressure — installed at the compressor discharge as a pressure relief

Q16. **A technician is called to a system with oil logging (oil trapped in the evaporator). What causes oil logging and how is it corrected?**

A) Oil logging occurs only in systems with TXV valves — replace with cap tube to correct

B) Oil leaving the compressor is not returning — low suction line velocity or wrong oil type

C) Oil logging is caused by excessive oil charge — drain excess oil from the system and recheck performance

D) Oil logging is caused by water contamination mixing with the oil — replace filter drier and all oil

Q17. **A reciprocating compressor has a broken valve reed. How would this affect compressor performance and what test would confirm it?**

A) Broken valve reed raises head pressure — confirmed by discharge pressure higher than normal

B) Broken reed allows gas to bypass compression — confirmed by a very low compression ratio

C) Broken valve reed causes the compressor to short-cycle — confirmed by rapid on/off cycling

D) Broken valve reed only affects oil pressure — confirmed by low oil pressure switch tripping

Q18. **A large commercial HVAC-R system is equipped with a demand defrost control (electronic defrost timer/controller). How does demand defrost differ from time-initiated/time-terminated defrost?**

A) Demand defrost runs the defrost heater at reduced power continuously to prevent frost — the heater never fully shuts off

B) Demand defrost uses a manual initiation only — an operator must start each defrost cycle

C) Demand defrost monitors actual frost accumulation and defrosts only when buildup requires it

D) Demand defrost and time-initiated defrost are functionally identical — the term "demand" refers to on-demand manual initiation

Q19. **Why has R-22 refrigerant been phased out in Canada?**

A) It is too expensive to produce

B) It is not efficient enough for modern systems

C) It reacts with copper tubing

D) It is an ozone-depleting HCFC

Q20. **The sight glass in a refrigeration liquid line shows bubbles continuously while the system runs normally. This indicates:**

A) The condenser is working too efficiently

B) Insufficient subcooling causing flash gas

C) Normal operation — bubbles are expected in the liquid line

D) Air in the refrigerant

Q21. **A scroll compressor makes a loud noise and vibrates as it starts, and the noise clears within a few seconds. What is the MOST likely cause?**

A) Reverse rotation from two swapped supply phases

B) A worn discharge valve letting gas blow back at start

C) A low refrigerant charge overheating the windings

D) Refrigerant that migrated into the shell while off

Q22. **The refrigerant R-600a (isobutane) is used in domestic refrigerators. What special precaution is required when servicing these units?**

A) It is a flammable A3 hydrocarbon — no ignition sources, rated recovery equipment

B) R-600a requires POE oil, and beyond that oil change no precautions are needed

C) R-600a runs at very high pressure, so ordinary service gauges cannot be connected

D) R-600a has a very high GWP, so the charge must go into a high-pressure cylinder

Q23. **What is the purpose of a suction line accumulator in a refrigeration system?**

A) To subcool liquid refrigerant before the metering device

B) To store additional refrigerant charge for use during peak demand

C) To catch liquid refrigerant and oil before they reach the compressor

D) To separate oil from the discharge gas before it enters the condenser

Q24. **An air-to-air heat pump is calling for heat but suction pressure is low and the technician suspects the reversing valve never shifted out of the cooling position. Which observation CONFIRMS the valve is still in cooling?**

A) Discharge pressure at the compressor reads higher than suction pressure

B) The reversing valve solenoid coil shows continuity with the power off

C) The outdoor coil is frosting while the indoor coil delivers warm supply air

D) The outdoor coil runs warm while the indoor coil blows cool air

Q25. **A large commercial refrigeration system experiences repeated compressor failures within months of replacement. No obvious cause is found after each failure. What systematic approach should be taken to find the root cause?**

A) Increase the frequency of filter drier replacement to monthly to prevent any contamination

B) Switch to a different refrigerant type as the current refrigerant may be incompatible with the compressor

C) Continue replacing compressors with higher-quality units until the problem stops

D) Perform a comprehensive audit of electrical, refrigerant, and operating conditions

Q26. **What is the purpose of the condenser in a refrigeration system?**

A) To compress refrigerant vapour for circulation

B) To reject heat and condense the refrigerant to a liquid

C) To evaporate the refrigerant using heat from the room

D) To regulate refrigerant flow to the evaporator

Q27. **A commercial refrigeration system uses a compressor with an economizer port. What is the purpose of the economizer circuit connected to that port?**

A) To reduce electricity consumption by cycling the compressor off during low-load periods

B) To subcool liquid refrigerant using refrigerant expanded at an intermediate pressure stage

C) To recover condenser heat for domestic hot water heating

D) To use outdoor air for cooling when ambient temperature is below setpoint, bypassing the refrigerant circuit

Q28. **A technician needs a cylinder to recover refrigerant from a system in Canada. Which cylinder is acceptable?**

A) A shop-built vessel fitted with a relief valve and a pressure gauge

B) A cylinder marked to a Transport Canada specification and in date

C) Any cylinder rated above the refrigerant's saturation pressure

D) The disposable cylinder the new refrigerant charge was supplied in

Q29. **R-410A operates at significantly higher pressures than R-22. What is the approximate high-side pressure of R-410A at 35°C condensing temperature?**

A) 380 psi

B) 450 psi

C) 290 psi

D) 175 psi

Q30. **A refrigeration system fitted with a thermostatic expansion valve is running with normal suction pressure, discharge pressure well above normal for the outdoor temperature, and compressor amp draw above normal. The MOST likely cause is:**

A) Worn or leaking compressor valves

B) An undercharge of refrigerant

C) Poor condenser heat rejection

D) The expansion valve stuck closed

Q31. **An electronic expansion valve (EEV or EXV) offers advantages over a TXV because:**

A) It is completely self-contained and never needs calibration

B) It sets a fixed opening at start-up and needs no superheat feedback

C) It requires no electrical power and operates purely on refrigerant pressure

D) It meters flow from superheat computed by its controller

Q32. **A walk-in freezer evaporator on electric defrost initiates every 6 hours and terminates either on its termination thermostat, set for this coil at 10°C (50°F), or on a 30-minute time limit. Every cycle runs the full 30 minutes and ends on the timer, yet when the technician opens the box at the end of a cycle the coil is completely clear of frost and the drain pan is hot. This indicates:**

A) Normal operation — defrost cycles run a fixed length regardless of need

B) The defrost heaters are undersized for the coil and its load

C) The termination thermostat is faulty or positioned incorrectly

D) Refrigerant undercharge is causing excess frost accumulation

Q33. **Warm humid air passes over an evaporator coil and water collects in the drain pan. Why does the coil take moisture out of the air?**

A) The refrigerant absorbs the water vapour and carries it outdoors

B) The low pressure inside the coil draws water vapour from the airstream

C) The drain pan draws moisture out of the air that passes above it

D) The coil surface is below the dew point, so moisture condenses

Q34. **A three-phase 460 V condenser fan motor is to be run from a pulse-width-modulated (PWM) variable frequency drive. Which arrangement puts the motor winding insulation most at risk from reflected-wave voltage spikes?**

A) An inverter-duty motor with a line reactor on the supply side of the drive

B) A general-purpose motor fed from the drive through a long motor cable

C) An inverter-duty motor fed through a dV/dt filter at the drive output

D) A general-purpose motor with the drive mounted beside it on short leads

Q35. **A medium-temperature refrigeration system is running on a call for cooling, with the box still well above setpoint, and the suction gauge reads below atmospheric pressure (a vacuum). This indicates:**

A) Discharge valves leaking inside the compressor

B) The system has been overcharged with refrigerant

C) A severe undercharge or a blocked metering device

D) Pump-down mode with the solenoid valve closed

Q36. **A refrigeration system has higher-than-normal suction pressure AND higher-than-normal discharge pressure simultaneously. What does this pattern most likely indicate?**

A) Refrigerant overcharge or poor condenser heat rejection (dirty coil, failed fan, or high ambient)

B) Refrigerant undercharge — low charge causes both pressures to be low

C) A restricted liquid line — restriction increases discharge pressure and starves the evaporator

D) Loss of compressor efficiency — worn valves make both pressures converge toward the middle

Q37. **Why must polyol ester (POE) oil be kept in a sealed container and its open time on site kept short?**

A) POE stops mixing with refrigerant once exposed to direct daylight

B) POE thickens on contact with air and will no longer reach bearings

C) POE evaporates once opened, so the container loses usable volume

D) POE pulls moisture out of the air quickly and then holds onto it

Q38. **After a compressor burnout, what contamination must be addressed before installing a replacement compressor, and what is the procedure?**

A) Acid, carbon, and moisture must be removed: flush, burnout driers, oil acid check, deep evacuation

B) Only the refrigerant oil needs replacement — flush the system with new refrigerant twice and recharge

C) Install the new compressor and run it until the contamination is naturally removed through normal operation over several weeks

D) Only the condenser and evaporator need flushing — the compressor discharge and suction lines self-clean during operation

Q39. **When recovering refrigerant from a system with a compressor that has burned out internally, what special procedure is required?**

A) Flush the system with nitrogen before recovering to dilute the acid

B) Use dedicated recovery equipment for acid-contaminated refrigerant and send it for reclamation

C) Standard recovery with a standard recovery machine — a burnt compressor does not affect the recovery procedure

D) Vent the refrigerant to atmosphere — contaminated refrigerant cannot be safely recovered

Q40. **A walk-in cooler compressor trips repeatedly on the low-pressure cutout (LPC) during normal winter operation, yet the cooler is maintaining setpoint temperature. What is the MOST LIKELY cause and the correct remedy?**

A) Low ambient is dropping head pressure and starving the TXV — install a head pressure control

B) The LPC setpoint is factory-set too high — lower the cut-out in the field to stop the nuisance trips

C) A restricted liquid line filter drier is reducing refrigerant flow and causing low suction pressure — replace the drier

D) Refrigerant overcharge is flooding the evaporator and lowering suction pressure — recover excess refrigerant

Q41. **In the refrigeration cycle, what state (phase) is the refrigerant as it leaves the condenser under normal operating conditions?**

A) A saturated mixture of liquid and vapour at the condensing temperature

B) Supercritical fluid, past the critical point and no longer condensable

C) Subcooled liquid, cooled below its condensing saturation temperature

D) Superheated vapour, still well above its own saturation temperature

Q42. **An EPR (Evaporator Pressure Regulator) valve is installed in the suction line of a multi-evaporator system. Its function is to:**

A) Maintain a minimum evaporator pressure setpoint

B) Limit maximum evaporator pressure to prevent overloading the compressor

C) Regulate head pressure at the condenser outlet

D) Balance refrigerant flow between multiple evaporators

Q43. **A liquid receiver on a supermarket rack is fitted with a pressure-relief valve. Under CSA B52, the relief discharge must be:**

A) Vented into the machinery room at low level so the vapour can be exhausted

B) Piped back into the suction line so the refrigerant is not lost

C) Left open at the valve outlet so that a discharge can be seen and heard

D) Piped to the outdoors, clear of building openings and air intakes

Q44. **A hermetic compressor has a locked rotor condition. The FIRST check before replacement is:**

A) Replace the compressor immediately — locked rotor is always a mechanical failure

B) Recharge the system with refrigerant

C) Check for liquid flood-back and verify supply voltage

D) Check the thermostat setpoint

Q45. **Why must a recovery cylinder never be filled with liquid refrigerant to the top of its internal volume?**

A) The recovery machine cannot pull a vacuum on a cylinder holding liquid

B) A blend separates into its components once the vapour space is gone

C) The cylinder valve passes vapour only and will seize on liquid flow

D) Trapped liquid has no room to expand when warmed and can burst it

Q46. **A technician is checking evaporator superheat on a system charged with R-407C. The gauge reads 40 psig at the evaporator outlet and the outlet line temperature is 32°F. The manufacturer's pressure-based PT chart lists three temperatures at 40 psig: Avg 16.2°F, Bubble 10.4°F and Dew 21.9°F. Which chart value gives the correct superheat?**

A) The bubble value, for a superheat of about 22°F

B) The liquid line temperature at the condenser outlet

C) The dew value, for a superheat of about 10°F

D) The average value, for a superheat of about 16°F

Q47. **A window air conditioner is running but not cooling. The evaporator coil is completely covered in ice. What is the MOST LIKELY cause?**

A) The refrigerant charge is too high, causing liquid to flood the evaporator

B) The condenser is too small for the application, reducing heat rejection

C) A weak run capacitor is making the compressor run slowly and overcool the coil

D) Restricted airflow over the evaporator dropping the coil below freezing

Q48. **A technician checks the "condenser split" on an air-cooled condensing unit. The condensing saturation temperature is compared against which temperature?**

A) The liquid line temperature leaving the condenser outlet

B) The return air temperature entering the evaporator coil

C) The dry-bulb temperature of the air entering the condenser

D) The wet-bulb temperature of the air leaving the condenser

Q49. **A commercial refrigeration system uses a floating head pressure control strategy. What does this mean and what is its benefit?**

A) The discharge pressure is fixed at the minimum possible setpoint regardless of ambient temperature

B) Head pressure fluctuates freely without controls, reducing wear on condenser fan motors

C) Condenser fans are controlled to let head pressure drop as low as ambient allows

D) The head pressure is allowed to rise and fall with refrigerant charge level, reducing the need for precise charging

Q50. **A hot gas defrost system is used in commercial refrigeration. How does it work?**

A) Electric heating elements mounted on the evaporator fins are energized during defrost

B) The refrigerant charge is partially recovered into a storage tank, allowing the evaporator to warm naturally

C) Hot compressor discharge gas is routed through the evaporator to melt frost

D) Hot ambient air is blown over the evaporator to melt frost

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 115-question 313A bank; the last page shows what the complete printable bank contains.

Q1. Answer: D

LPC tripping = suction pressure falling below cut-out. Low refrigerant charge reduces suction pressure below the cut-out setpoint. Restricted liquid line, dirty evaporator coil, or low evaporator load can also cause low suction. The LPC protects the system from operating at pressures where evaporator freeze-up or compressor damage could occur.

Q2. Answer: D

On a privately owned commercial system on private land, the halocarbon rules that bind the technician are the province's or territory's own, not the federal ones. The Federal Halocarbon Regulations, 2022 (SOR/2022-110), made under the Canadian Environmental Protection Act, 1999, open with an application section: they apply to air-conditioning or refrigeration systems, solvent systems, fire-extinguishing systems and containers located in Canada that are owned by Her Majesty in right of Canada, a board or agency of the Government of Canada, a Crown corporation as defined in the Financial Administration Act, or a federal work or undertaking, or that are located on aboriginal lands or federal lands. A supermarket owned by a private grocery chain on private land is none of those, so that instrument does not reach this job.

Inside its own scope the federal regulation is strict, and it is worth knowing in its own terms because federal sites do come up. It provides that a person must not release, or allow or cause the release of, a halocarbon contained in an air-conditioning or refrigeration system, a fire-extinguishing system or a container, or in equipment used in the reuse, recycling, reclamation or storage of a halocarbon. The prohibition does not apply if the release is for the purpose of calibrating leak-detecting devices with equipment designed specifically for that purpose and the manufacturer's recommended procedures are followed; results from connecting or disconnecting hoses that are less than 1 m in length and used for charging with, or recovering, a halocarbon; results from a purge system on an air-conditioning or refrigeration system, including any associated recovery equipment, that emits less than 0.1 kg of halocarbon per kilogram of air purged; or is from a fire-extinguishing system for the purpose of fighting a fire that is not set for training purposes, or of testing the system in a military vehicle as authorized by a permit. Note the boundaries as written: a hose of exactly one metre is not inside the hose exception, and a leak-detector calibration is only inside its exception when both conditions, the purpose-built equipment and the manufacturer's procedure, are met. The regulation also provides that only a certified person may install, service or recover a halocarbon from an air-conditioning system or refrigeration system. None of that changes who the regulation applies to.

Outside that scope, Environment and Climate Change Canada describes provincial and territorial legislation on ozone-depleting substances and halocarbon alternatives as complementing the federal regulations, and lists among its requirements proper labelling and handling of equipment, training for equipment service providers, procedures for installing, removing, servicing, repairing or decommissioning products, and a prohibition on recharging leaking equipment. So the rules that actually bind this job, meaning who may handle the refrigerant, what has to happen to it instead of a release, and what has to be recorded, come from the halocarbon regulation of the province or territory where the work is done, and they differ in detail from one province to the next.

The other two instruments are federal and are real, which is what makes them tempting. The Ozone-depleting Substances and Halocarbon Alternatives Regulations control the manufacture, import, export, sale and use of the substances themselves, through prohibitions, permits and allowances; they contain no release prohibition for a refrigeration system and say nothing about who may service one. The Transportation of Dangerous Goods Regulations do bear on this trade, but on the recovery cylinder, its TC specification and its movement on the road, not on how the system is serviced.

Q3. Answer: B

Boiling absorbs latent heat, and that is far more heat per kilogram than simply warming the refrigerant. Heat that turns a liquid into a vapour is latent heat, and for a refrigerant it is many times the sensible heat picked up by warming that same mass a few degrees. The evaporator is therefore built to boil the refrigerant off, and only the last stretch of the coil adds sensible heat as superheat. That is why a modest mass flow can carry a large cooling capacity, and why flooding or starving the coil, which changes how much of the coil is actually boiling, costs capacity so quickly. It is also why the refrigerant leaving the coil must be all vapour: any liquid still in it represents latent heat the coil failed to use, and it is heading for the compressor.

Q4. Answer: C

Voltage only tells the truth under load, and a drop measured between two points is made in the length of circuit between them. Standing still, a corroded lug, a loose terminal or a burnt contactor contact carries no current and drops no voltage, so everything measures perfectly. Ask the compressor to start and it draws locked-rotor current for a moment; that current through a high-resistance joint produces a large voltage drop, and what reaches the motor is no longer enough to develop starting torque. The rotor stays put, locked-rotor current continues, and the overload does its job. Because the two readings were taken at the same instant, they localize the fault: full voltage at the disconnect with low voltage at the terminals puts the loss between them - pitted contactor contacts, a loose or corroded connection, an undersized or badly spliced conductor. Look for discolouration and heat damage, and measure the drop across each connection and across the closed contactor while the machine is trying to start.

If the supply itself were weak, the disconnect reading would sag along with the terminal reading instead of holding at 240 V. A failed start capacitor leaves full voltage standing at the terminals while the motor hums, so it puts no drop between the two measuring points. A seized compressor also draws locked-rotor current, but through a sound supply path the terminal voltage would stay close to the disconnect reading rather than falling 50 volts below it.

Q5. Answer: C

The 700 series is reserved for inorganic refrigerants, and the last two digits are the substance's molecular mass. Ammonia has a molecular mass of 17, so it is R-717; carbon dioxide is 44, so it is R-744; water is 18, so it is R-718; air is taken as 29, so it is R-729. The series is worth knowing because a number tells you immediately what family you are dealing with. The 400 series is zeotropic blends and the 500 series azeotropic blends, and both are numbered in the order they were registered rather than by what is in them — which is why R-407A and R-407C hold the same components in different proportions, are different refrigerants, and are not interchangeable. The 600 series covers organic compounds, including R-600a, isobutane. Halocarbons take their numbers from their chemical formula.

What a refrigerant number never encodes is a pressure rating or a safety property. Toxicity and flammability come from the separate ASHRAE safety group — the letter-and-number code such as A1, A2L or A3 — and pressure comes from the refrigerant's own saturation curve, which is why the pressure-temperature chart exists.

Q6. Answer: C

Three distinct terms: Recovery (remove to cylinder), Recycling (on-site oil/moisture cleanup), Reclaiming (certified facility to AHRI 700 virgin purity). Recovery: extracting refrigerant from a system into an approved recovery cylinder — mandatory before opening. Recycling: passing recovered refrigerant through filters and oil separators for reuse on the SAME system or same owner's equipment. Reclaiming: sending to a certified reclaim facility for processing to AHRI Standard 700 purity (essentially virgin quality, sold for any use).

Q7. Answer: C

A phase monitor relay watches the three-phase supply and opens its contacts, dropping the control circuit, when the supply is not fit to run a motor: a lost phase, two phases reversed, or phases out of balance. Lennox describes its A42 kit as a three-phase line monitor that protects against phase loss, phase reversal and phase unbalance, with contacts that open immediately if a phase is lost; Macromatic's monitor relays add undervoltage and overvoltage to the list.

Phase loss: Macromatic notes that a motor that has lost a phase draws locked-rotor current during start-up, and that a three-phase motor will continue to run after losing a phase, resulting in possible motor burn-out; US Motors (Nidec) adds that the motor then runs as a single-phase machine, with the current in the phases still connected greater than the winding was designed for. Phase reversal: reversing any two of the three phases makes a three-phase motor run in the opposite direction (Macromatic). A scroll compressor compresses in one rotational direction only; run backwards it builds no pressure differential and may make an abnormal sound. Copeland states that running a three-phase scroll in reverse for a short period (under one hour) has no negative impact on durability, that its internal overload trips after several minutes of reverse running, and that repeated restarts in reverse permanently damage the bearings through oil loss; some Copeland models carry an electronic module that will not let the compressor run when the phasing is wrong. Bitzer's screw compressor instructions warn to operate the compressor only in the intended rotation direction, and Lennox calls its A42 ideally suited to protect scroll and screw compressors from reverse rotation. Phase unbalance: unequal phase voltages produce running currents unbalanced by roughly 6 to 10 times the voltage unbalance (US Motors, based on NEMA MG 1).

The relay is a protective device, not a power supply. It cannot make a third phase out of a single-phase service, it does not move load between the phases to save energy, and it monitors only the electrical supply, so it has no way to know whether the system is undercharged. Confirm rotation after any supply work upstream of the unit: suction pressure should drop and discharge pressure rise when the compressor starts (Copeland).

Q8. Answer: A

Letting the nitrogen out is not the same as taking it out. When the gauges read zero the system is still full of nitrogen at atmospheric pressure, and only a vacuum pump removes it. The sequence after a nitrogen test is release, evacuate, then charge; this technician went from release straight to charging, so the refrigerant was weighed in on top of a system full of nitrogen. Nitrogen does not condense at any pressure or temperature the condenser can reach, so it is carried round to the condenser and stays there, occupying surface that should be rejecting heat and adding its own partial pressure to the refrigerant's. Head pressure climbs beyond anything the ambient and the load can justify, the high-pressure control trips, the unit restarts when the pressure falls back and trips again, and the little running it does is at reduced capacity. There is no bleeding it off: recover the charge, evacuate to a deep vacuum proven with a micron gauge on a standing test, and weigh the charge back in.

The other faults are real but each writes a different signature. A plugged filter drier is a liquid line restriction: it starves the evaporator, so suction pressure runs low with high superheat and the drier outlet runs colder than its inlet; head pressure does not climb and the high-pressure control has no reason to trip. Moisture freezing at the metering device also starves the coil, intermittently, with the same low-side picture and a moisture indicator turning colour; again nothing pushes the high side up. Compressor valve damage is the opposite of what was found: a compressor with damaged valves still turns and moves some gas but pumps poorly, so suction pressure runs higher than it should, discharge pressure lower, the two converge and amp draw falls, which is a machine that cannot reach its high-pressure cut-out rather than one tripping it repeatedly.

Q9. Answer: A

Resistance between terminals proves the windings are continuous; only an insulation test says whether they are still isolated from the shell. Common-to-start and common-to-run readings that sit in spec confirm the copper is intact and the windings are balanced, which is exactly why the meter looks reassuring on a motor that is finished. An insulation resistance tester applies a high test voltage between the windings and the earthed shell, and a very low reading means the insulation has broken down and current is leaking to ground. That is a grounded motor: it cannot be repaired in the field, and the compressor is replaced. Treat it as a burnout rather than a simple swap, because the same event that destroyed the insulation loads the circuit with acid and carbon. Check the oil for acid, clean the circuit, fit the appropriate driers and evacuate properly, or the replacement will fail the same way. For contrast, an open internal overload reads as an open circuit between terminals rather than as a path to ground, and a hermetic motor is never intended to conduct through its shell. Isolate, lock out and confirm the circuit is dead before applying an insulation tester, and discharge the windings afterwards.

Q10. Answer: B

Indoor running but outdoor unit not starting: check power supply to the outdoor unit first. The indoor blower fan and outdoor compressor/condenser fan run on separate circuits. A tripped circuit breaker, open disconnect switch, or blown fuse at the outdoor unit disconnect kills the outdoor circuit while leaving indoor running. After confirming power: check the contactor (coil energized? contacts closed?), capacitor (start/run capacitors fail frequently — multimeter capacitance test), and check for safety control lockout (high/low pressure switch).

Q11. Answer: B

Metering device (TXV or fixed orifice): creates the pressure drop separating the high and low sides of the system. The TXV (Thermostatic Expansion Valve) or fixed orifice/cap tube meters refrigerant flow, dropping pressure rapidly. This pressure drop causes partial flashing (evaporation), dramatically lowering refrigerant temperature. The resulting cold liquid/vapor mixture then enters the evaporator where it absorbs heat and fully evaporates.

Q12. Answer: A

R-454B (Opteon XL41) is an A2L blend of R-32 and the hydrofluoroolefin R-1234yf, 68.9/31.1 by weight, that Chemours offers to replace R-410A in new equipment designs. Chemours describes it as a mildly flammable, low-GWP HFO-based refrigerant - a direct replacement, new equipment only, for all equipment types designed for R-410A - with similar properties to R-410A and a GWP of 466 (AR4) against R-410A's 2088 (AR4), a 78% reduction. R-32 on its own is also used in place of R-410A: Daikin uses it in its ductless mini-splits, air conditioners and heat pump systems, puts its GWP at 675 against R-410A's 2,088 and calls the transition from the blended R-410A to the single-component R-32 essential; Canada's Ozone-depleting Substances and Halocarbon Alternatives Regulations (SOR/2016-137) list HFC-32 at a GWP of 675 in Table 4 of Schedule 1. Both are safety class A2L, lower flammability - R-454B under ISO/ASHRAE 34 (Chemours), R-32 under ISO 817:2014 (Daikin) - so Chemours directs users to the local regulations and standards (UL, IEC, ASHRAE) for the allowable system charge, new equipment design and safe handling, and notes that XL41 is miscible with POE lubricant.

Reading the wrong answers: Chemours states that, at the time of its product information, there are currently no comparable non-flammable, low-GWP alternatives with pressures close to those of R-410A, so carbon dioxide is not one; R-125 is a single-component HFC, but its GWP is 3 500 (SOR/2016-137, Table 4), above R-410A's, not below; and R-22 is an HCFC - the same Regulations prohibit manufacturing, and with narrow exceptions importing, a product containing or designed to contain HCFC-22 - so nothing is bringing it back for new equipment, whatever its GWP.

Q13. Answer: D

High-pressure cutout: safety device that shuts down compressor at excessive head pressure. High head pressure can be caused by: dirty/blocked condenser, condenser fan failure, refrigerant overcharge, non-condensables in the system, or high ambient temperature. If head pressure reaches the cutout setpoint (typically 590–650 PSI for R-410A systems), the compressor is de-energized — protecting the compressor, condenser, and refrigerant circuit from over-pressure damage. The manual reset type requires a technician reset, preventing restart until the cause is investigated.

Q14. Answer: A

HFC refrigerants require synthetic POE (polyolester) oil for proper lubrication and oil return — mineral oil is not miscible and will cause oil logging and lubrication failure. R-12 and R-22 (CFC/HCFC) systems use mineral oil or alkylbenzene oil. HFCs (R-134a, R-404A, R-410A, R-407C) require POE oil, which mixes with HFCs so the oil is carried around the circuit and returned to the compressor. Mixing mineral oil with HFCs causes the oil to separate out and pool in the evaporator, starving the compressor. POE oil is strongly hygroscopic — it takes up atmospheric moisture quickly and holds it, so containers stay sealed and open time is kept to a minimum.

Q15. Answer: C

Check valve: allows flow in one direction only — prevents refrigerant flowing backward during the off-cycle. Used in discharge lines on heat pumps and tandem/parallel compressor systems. On heat pumps: discharge line check valves direct refrigerant correctly when the reversing valve switches modes. On parallel/tandem compressor systems: individual discharge line check valves prevent the running compressor from pushing hot discharge gas into the stopped compressor. Also used in liquid lines of multi-evaporator systems to prevent migration.

Q16. Answer: B

Oil logging: oil circulates out of the compressor but cannot return — inadequate oil circulation lets it accumulate in low points and the evaporator. Refrigerant velocity carries oil through the system. If suction line velocity is too low (system running at partial load, oversized or incorrectly sized/routed piping), oil drops out of the refrigerant stream and pools. Risers need oil traps (P-traps) sized to maintain velocity even at minimum load. Wrong oil type/viscosity: too thick to flow at evaporator temperature. Low refrigerant charge also reduces velocity. Correction: verify line sizing and re-pipe as needed, add traps/risers, correct refrigerant velocity, verify oil specification and compatibility.

Q17. Answer: B

Broken compressor valve reed: gas bypasses the damaged valve instead of being compressed, severely reducing compression efficiency and capacity. A broken suction reed lets cylinder gas escape back into the suction line on the compression stroke — suction pressure runs high because the compressor can no longer pull it down, discharge pressure runs low, and mass flow drops. A broken discharge reed lets compressed gas leak back into the cylinder on the suction stroke and re-expand — again suction pressure rises and discharge pressure falls. Either way the two pressures converge, so a fault that pushes head pressure up is not this signature. Amp draw is below normal because the compressor is doing less work, while discharge temperature climbs because the same gas is compressed over and over. Confirm by comparing the measured compression ratio against the ratio the system's design evaporating and condensing conditions should produce — there is no single healthy number, since a low-temperature system normally runs a far higher ratio than a comfort-cooling system. A valved-out pump-down check (front-seat the suction service valve and watch whether the compressor pulls the low side down and holds it) separates a worn compressor from a system fault. A reed failure does not by itself cycle the compressor on and off, and it is not confined to the oil circuit — it shows in the refrigerant-side pressures long before an oil pressure switch reacts. Verify with valve plate inspection before condemning the compressor.

Q18. Answer: C

Demand defrost: defrosts only when needed (based on measured frost accumulation) vs fixed-time schedule. Time-initiated/time-terminated defrost runs defrosts at fixed intervals (e.g., every 6 hours) regardless of frost buildup — wastes energy when little frost is present, and may not defrost often enough under heavy conditions. Demand defrost senses frost via air pressure differential across the coil, a capacitance sensor, or temperature-rise sensing, and initiates defrost only when buildup is sufficient. Benefits: fewer unnecessary defrosts and energy savings (up to 30% reduction in defrost energy), improved product quality (less product temperature cycling), longer equipment life.

Q19. Answer: D

R-22 (HCFC) = ozone depleter. Because of its ozone-depleting potential, Canada phased out R-22 production and import under the Montreal Protocol. As of 2020, R-22 can only be used from recovered/recycled stock. Common replacements include R-410A (new systems) and R-407C (retrofit).

Q20. Answer: B

Bubbles in sight glass = flash gas = subcooling problem. Liquid refrigerant should arrive at the metering device as a solid liquid (no bubbles). Bubbles (flash gas) indicate: refrigerant is partially vaporizing before the metering device — caused by low charge (low subcooling), restriction in liquid line (pressure drop = boiling), or high liquid line temperature.

Q21. Answer: D

Noise and vibration that last only a moment at start-up point to refrigerant that migrated into the compressor during the off cycle. Copeland's application bulletins for its air-conditioning scrolls warn that excessive migration of refrigerant to the compressor during the off cycle can cause oil pump-out on start-up, excessive starting noise and vibration, and bearing erosion, and Carrier's service manual reads straining or vibration at start-up that clears quickly as a sign of an off-cycle migration problem. The scroll itself tolerates some liquid - Copeland's compliance design lets the scrolls separate so liquid or debris can pass without damaging the compressor - but that is tolerance, not immunity: excessive floodback or repeated flooded starts dilute the oil and wear the bearings, and Copeland adds broken scrolls if the hydraulic slugging pressure is high enough. Copeland's protection is a crankcase heater, which keeps refrigerant out of the shell during standstill. Copeland ties the crankcase heater to the compressor's charge limit, and the rule depends on the model: on its 1.5 to 5 ton air-conditioning scrolls a heater is recommended on single-phase models when the system charge is over the limit and required above 120% of it, and it is required on three-phase models when the charge exceeds the limit and an accumulator cannot be piped to drain liquid freely during the off cycle; on its three-phase ZBKQ refrigeration scrolls it is required whenever the system charge exceeds the charge limit. For its 1.5 to 5 ton air-conditioning scrolls Copeland does not recommend a pump-down cycle for migration control. The other choices do not fit a noise that clears in seconds: reverse rotation gives no pressure differential and may sound abnormal for as long as the compressor runs backwards; a leaking discharge valve shows up after shutdown, not at start-up - the scroll's discharge check valve exists to stop high-side gas flowing rapidly back through the compressor after shutdown, and the reverse-rotation sound Copeland's shut-down devices deal with comes as high-pressure gas equalizes back through the scrolls at shutdown; and Danfoss ties a low charge to higher superheat, lower subcooling and lower evaporator pressure, not to a start-up noise.

Q22. Answer: A

R-600a (isobutane) is a highly flammable A3 refrigerant — flammability is the primary hazard. Hydrocarbon refrigerants (R-600a, R-290/propane) have excellent thermodynamic properties but they burn. They are used in small-charge domestic appliances, where the charge is small enough that a full leak stays below the lower flammability limit in the room. Servicing: ventilated areas only, no open flames or sparks, non-sparking tools, and recovery equipment rated for flammable gases. These appliances must not be serviced in an enclosed space. R-600a is a low-GWP refrigerant and it operates at lower pressures than the common HFCs, so neither a GWP argument nor a high-pressure argument is what drives the precautions.

Q23. Answer: C

Suction accumulator: liquid trap in suction line that holds back liquid refrigerant and oil that might otherwise enter and damage the compressor (liquid slugging). Positioned between the evaporator and compressor, it allows vapor to pass through while holding back liquid. A small metered orifice slowly returns trapped oil back to the compressor. Common on heat pumps (liquid can return during defrost) and systems with large refrigerant charges. Oil trapped too long = compressor oil starvation.

Q24. Answer: D

Which coil is rejecting heat tells you which way the valve is set. In heating, the reversing valve sends discharge gas to the indoor coil, so the indoor coil is the condenser (hot, warm supply air) and the outdoor coil is the evaporator (cold, often frosted). If the valve is stuck in the cooling position during a call for heat, those roles are swapped: the outdoor coil runs warm because it is still acting as the condenser, and the indoor coil is boiling refrigerant, so the supply air comes off cool. That reversal is the confirmation.

A frosted outdoor coil with warm supply air is what correct heating operation looks like, not a fault. Discharge pressure being higher than suction pressure is true of every running system in either mode, so it distinguishes nothing. Continuity through the solenoid coil only proves the winding is not open — the coil can read fine while the valve body is mechanically stuck or while the coil is never energized by the control circuit, so it must be backed up by measuring voltage at the coil during the heat call and by feeling the valve body tubes.

Q25. Answer: D

Repeated compressor failure: symptom of an unresolved underlying problem — requires systematic root cause analysis. Audit: 1) Electrical: voltage quality (sags/harmonics), single-phasing, phase imbalance, improper motor protection settings. 2) Refrigerant circuit: operating pressures vs design (high head/low suction = high compression ratio), superheat/subcooling, liquid slugging (low superheat, flooded start), contamination and oil acid levels (acid, moisture). 3) Lubrication: wrong oil, oil logging, oil returning contaminated. 4) Operating conditions: high ambient, overloaded system, defrost effectiveness. 5) Installation: improper piping, incorrect electrical protection. Review operating logs for patterns, and inspect each failed compressor (failure mode determines root cause).

Q26. Answer: B

Condenser: rejects heat, converts vapour to liquid. High-temperature, high-pressure vapour from the compressor enters the condenser. Heat is rejected to the outside environment — ambient air (air-cooled) or water (water-cooled). As heat leaves, the vapour condenses into a high-pressure liquid ready for the metering device.

Q27. Answer: B

Refrigerant economizer: improves system efficiency and capacity by subcooling liquid using a flash tank or heat exchanger at intermediate pressure. A portion of liquid refrigerant is expanded to an intermediate pressure in the economizer. This cold refrigerant subcools the main liquid line, reducing flash losses at the metering device and increasing refrigerating effect per kilogram circulated. The intermediate-pressure flash gas is injected into the compressor mid-stage through the economizer port (requires a two-stage or economizer-capable compressor). Because flash losses grow as the pressure ratio grows, the gain is largest on low-temperature duty and in high ambient conditions. An air-side economizer, which brings in outdoor air when it is cool enough to carry the load, is a separate device on the air side and does nothing to the refrigerant circuit. Condenser heat reclaim for domestic hot water and simply cycling the compressor off at low load are also separate strategies, not the function of the economizer port. Common in commercial refrigeration and VRF systems.

Q28. Answer: B

A cylinder that will hold recovered refrigerant is a pressure receptacle for the transport of dangerous goods, so it has to carry a specification Transport Canada authorizes and it has to be within its requalification date. Transport Canada's own container pages describe the permanent TC markings a specification cylinder carries under CSA B339, and the requalifier's registered mark and date stamped on the cylinder each time it is requalified; CSA B340 governs the selection and use of those containers. Cylinders sold here are frequently dual-marked, because Transport Canada also lists DOT cylinders manufactured to the American rules among the authorized container types — so 'DOT-rated' on its own is US shorthand, and what actually matters is a specification Transport Canada accepts, plus a current requalification date.

A single-use disposable cylinder is out on both counts: it was never built to be refilled, it has no overfill protection, and refilling it is prohibited. A shop-built vessel carries no specification marking at all, whatever fittings are hung on it. And a pressure rating is not a specification: a cylinder can be strong enough for the gas and still be unmarked, unauthorized or long out of date. Fill by weight against the cylinder's tare and rated capacity so a vapour space is left, and keep a separate, clearly labelled cylinder for a burnout or for mixed refrigerant.

Q29. Answer: C

R-410A at a 35°C (95°F) condensing temperature sits near 295 psig on the high side, so 290 psi is the closest choice. R-22 at that same 35°C sits near 182 psig, so the gap at condensing conditions is about 60 per cent. Lower down the curve the gap is wider: at a 40°F evaporator temperature R-22 is near 69 psig while R-410A is near 118 psig, about 70 per cent higher. There is no single multiplier that covers the whole curve, which is why any comparison of the two has to name the saturation temperature it is made at, and has to compare gauge with gauge — reading one refrigerant as gauge pressure and the other as absolute makes the gap look far smaller than it is. On a hot day the measured head pressure runs higher still, because the condensing temperature sits well above ambient. All of this is why R-410A systems require components, hoses, gauges and recovery equipment rated for the higher pressure.

Q30. Answer: C

Head pressure that has climbed while suction pressure stayed where it belongs says the condenser is not getting rid of its heat. A condenser can only reject heat into what it is given, so a fouled or blocked coil, a fan that has failed or is turning the wrong way, discharge air recirculating around the unit, a high ambient, or non-condensable gas occupying condenser volume all force the refrigerant to condense at a higher temperature than it should, and the pressure rises with that temperature. The compressor is then pumping against a wider difference, doing more work, which is exactly what the raised amp reading is reporting.

Suction pressure stays near normal on this machine because the thermostatic expansion valve throttles to hold superheat and goes on feeding the coil at the rate the load calls for. The same fault reads differently on a fixed-orifice machine: there the higher liquid pressure pushes more refrigerant through the orifice, so suction pressure rises along with the head and the pattern becomes high on both sides.

The other three faults move the pressures in other directions. An expansion valve stuck closed starves the coil, so suction pressure falls well below normal and discharge falls with it. Worn or leaking compressor valves let gas bleed back instead of being pumped, so the machine still turns but pumps poorly: suction pressure runs higher than it should because the compressor cannot pull it down, discharge pressure runs lower than normal because it cannot build head, the two pressures come closer together, and amp draw falls rather than rises. An undercharge leaves too little refrigerant in the system, so suction pressure, discharge pressure and subcooling all read low together.

Q31. Answer: D

EEV: electronic precision control based on multiple sensor inputs. The controller reads suction pressure from a transducer and suction line temperature from a sensor, calculates superheat from the two, and drives a stepper motor to reposition the valve — much faster response and a far wider control range than a mechanical TXV, which is why EEVs suit inverter-driven and VRF equipment. The other answers describe things an EEV is not. It is not a pressure-operated valve: without control power and a working controller it does not meter at all. It is not self-contained either — the valve is only one part of a package that includes the controller and its sensors, and the superheat target is a programmed setting that can be checked and adjusted. And it does not park at a fixed opening: superheat feedback is recalculated continuously, which is precisely the advantage being asked about.

Q32. Answer: C

A coil that comes out of defrost completely clear has had enough heat; if the cycle still ran to its time limit, the control never saw the coil warm up. Temperature termination is how an electric defrost is meant to end - once the ice is gone the coil metal climbs quickly past the termination setting and the cycle stops, which saves energy and stops the box temperature climbing any further. The 30-minute limit is a fail-safe, not the intended terminator. Here the ice is gone and the drain pan is hot, so heat input and frost load are not the problem; what has failed is the device that is supposed to notice. Either the termination thermostat has failed open or drifted out of calibration, or it is clamped where it never sees warm metal - on a return bend that stays cold, or away from the heater path. Test it for continuity as its temperature rises and confirm it is mounted where the equipment's installation instructions place it. Heaters that were undersized for the coil and its load would leave ice still on the coil when the timer ended, which is exactly what is not found here. An undercharge is a refrigeration-side fault and has no bearing on whether a defrost cycle terminates. And a defrost is not supposed to run a fixed length regardless of need - preventing that is precisely why temperature termination exists.

Q33. Answer: D

A cooling coil dehumidifies because its surface sits below the dew point of the air passing over it. Air holds water vapour; cool any surface below the temperature at which that vapour would begin to condense — the dew point — and water comes out of the air onto the surface, exactly as it does on a cold glass. On a coil it runs down the fins into the drain pan and leaves the building as condensate. Removing that moisture is latent cooling; lowering the air temperature is sensible cooling, and a coil does both at once, which is why a coil handling humid air spends part of its capacity on water rather than on temperature.

The refrigerant never touches the air — it is sealed inside the tube, so it can neither absorb the moisture nor be affected by it, and the low pressure inside that sealed circuit has no reach into the airstream. The drain pan only catches what has already condensed. Two practical consequences follow from the real mechanism: a coil running warmer than the dew point cools without dehumidifying, and a coil driven below freezing turns the condensate to ice and blocks its own airflow.

Q34. Answer: B

The arrangement at risk combines the two conditions behind reflected-wave damage: fast-switching PWM pulses sent down a long motor cable, and a general-purpose motor, whose peak-voltage limit is lower than that of an inverter-duty motor. When the motor impedance is greater than the impedance of the cable between motor and drive, the voltage waveform reflects at the motor terminals and builds a standing wave, and longer motor cables favour higher-amplitude standing waves. The spikes can break down the winding insulation and cause phase-to-phase or turn-to-turn short circuits (US Department of Energy, Motor Systems Tip Sheet 14, which cites NEMA MG 1-2011). NEMA MG 1-2011 Part 30 expects no significant reduction in the service life of a general-purpose motor on a drive when the peak voltage at its terminals is limited to 1,000 V and rise times are 2 microseconds or more; beyond that, the tip sheet sends the installer to the motor maker. Under Part 31, the insulation of a definite-purpose inverter-duty motor rated 600 V or less is designed to withstand peaks of up to 3.1 times its rated line-to-line voltage, about 1,426 V for a 460 V motor. Each of the other arrangements has a protection the same tip sheet lists: an inverter-duty motor, a dV/dt filter, or a short distance between drive and motor, since damaging reflected waves are generally not a problem when that distance is less than 15 ft (about 4.6 m). The tip sheet also notes that spikes on 208 V or 230 V systems are unlikely to damage insulation at any cable length, which is why the motor voltage matters. The same tip sheet says inverter-ready general-purpose motors are well suited to VFD control of variable torque loads, so a general-purpose motor on a fan is not wrong in itself: the risk is the peak voltage at its terminals. Where a long run cannot be avoided, obtain the inverter rise times and the effects of cable length from the drive and motor manufacturers.

Q35. Answer: C

A vacuum on the low side of a running system means almost nothing is reaching the evaporator. With the box calling for cooling, the compressor is pulling vapour out of the coil faster than the metering device can replace it, so the pressure falls until it drops below atmospheric. Two faults do that: a severe undercharge, where there is barely any refrigerant left to feed the coil, and a blockage that stops the flow - a metering device stuck closed, a plugged strainer or filter-drier, or a kinked liquid line. Either way the compressor loses the vapour flow that returns its oil and, on a suction-cooled motor, carries the winding heat away, so shut the system down and diagnose before it is damaged. Running below atmospheric pressure also draws air and moisture into the low side through any leak, fitting or seal, which puts non-condensables and acid into the system. The other readings do not fit. An overcharge raises suction pressure rather than lowering it. Leaking compressor discharge valves let high-side gas bleed back into the cylinder, which pushes suction pressure UP while head pressure falls toward it. And a pump-down draws the low side down only after the call for cooling has ended and the liquid-line solenoid has closed, not while the box is still calling.

Q36. Answer: A

High suction + high discharge simultaneously = refrigerant overcharge or a condenser that cannot reject heat. An overcharge backs liquid up in the condenser and raises both pressures; a dirty or blocked condenser, a failed condenser fan, or high ambient temperature prevents heat rejection with the same result. Check condenser cleanliness, fan operation, and refrigerant charge. Contrast this with worn compressor valves, where suction rises while discharge FALLS and the two pressures converge toward the middle — that is compressor inefficiency, not this pattern.

Q37. Answer: D

POE is hygroscopic: it absorbs atmospheric moisture far faster than mineral oil, and it holds that water chemically rather than simply carrying it along. That is what makes it a field problem. Free water in a system can be boiled off with a deep vacuum, but moisture bound into POE largely will not come out on the vacuum pump, so an open pail left on the truck for an afternoon can carry water straight into a circuit that was correctly evacuated. Water plus POE plus heat gives hydrolysis, hydrolysis gives acid, and acid gives copper plating, varnished valves and eventually a burnout. Working practice follows from that: buy oil in the smallest sensible container, keep the cap on until the moment of use, pour rather than leave it standing open, never return poured oil to the container, and fit a fresh liquid line drier whenever the circuit has been opened. POE is used in the first place because HFC and HFO refrigerants are not miscible with mineral oil, so this handling discipline is the price of that compatibility.

Q38. Answer: A

After a compressor burnout, acid, carbon, and moisture are spread through the whole system and all of it must be removed before the replacement compressor runs. Acid formed by the thermal breakdown of oil and refrigerant attacks the winding insulation and plates copper onto bearing and valve surfaces, so a system left dirty destroys the new compressor. Procedure: 1) Recover the refrigerant. 2) Clean the lines and coils — flush with an approved flushing agent or blow through with dry nitrogen to carry out acid and carbon. 3) Fit an oversized burnout filter drier in the liquid line and a suction line drier ahead of the compressor. 4) Install the new compressor and charge it with the oil specified for that machine. 5) Triple evacuate — pull a vacuum, break it with dry nitrogen, repeat, then pull the final vacuum and prove it with a decay test, watching that the reading holds steady instead of creeping back up. 6) Charge and run the system, then acid-test the oil after the first day or two of operation; if acid or discolouration remains, change the drier cores and retest, and check the system again about two weeks later.

Q39. Answer: B

Burned-out compressor: refrigerant is acid-contaminated — use dedicated equipment and send for reclamation. A burned motor creates acid and carbon deposits from the breakdown of motor windings (copper, varnish, refrigerant oil). This acid-contaminated refrigerant cannot be recycled back into a system — it must be sent to a reclamation facility (not just recycling). Use dedicated "burnout" recovery equipment to prevent contaminating your regular recovery machine, and clearly label the recovery cylinder as contaminated.

Q40. Answer: A

In cold weather the air passing over the condenser is colder, the condenser rejects heat too easily, and condensing (head) pressure falls. A thermostatic expansion valve is driven by the pressure difference across it, so it needs a minimum condensing pressure to push its rated flow. When head pressure sags, flow through the valve drops, the evaporator is starved, suction pressure falls below the low-pressure cut-out, and the compressor trips — even though the box is still holding setpoint, because the trips are short and the load is light in winter. The remedy is a head pressure control that holds condensing pressure up as ambient falls: condenser fan cycling on pressure or temperature, a fan speed controller, or a flooding valve that backs liquid into the condenser to reduce its active surface. The minimum condensing pressure it is set to hold is whatever keeps enough differential across the metering device for that refrigerant and that evaporator temperature, so it is taken from the system design data rather than from a rule of thumb. Lowering the low-pressure cut-out only hides the symptom: the evaporator stays starved, refrigerant velocity stays too low to return oil, and the compressor runs short of both cooling and lubrication. A restricted drier is a year-round restriction, not a seasonal one, and it shows up as a temperature drop across the drier with high superheat; an overcharge raises head pressure rather than lowering it.

Q41. Answer: C

Condenser outlet: subcooled liquid under normal operation. The condenser removes heat from the refrigerant in three stages: 1) desuperheating (vapour cools to saturation), 2) condensing (vapour → liquid at constant pressure and temperature), 3) subcooling (liquid cools further, below saturation). Subcooling is what stops vapour bubbles forming in the liquid line before the metering device. The subcooling to work to is the one the equipment manufacturer gives on the nameplate or in the installer's guide - Trane's TXV kit instructions, for example, tell the installer to charge to a minimum of 8°F subcooling or the outdoor unit nameplate subcooling, whichever is higher.

Q42. Answer: A

EPR valve: sets minimum evaporator pressure (warmer setpoint), preventing the evaporator from getting too cold. In multi-temperature systems, different evaporators need different temperatures. The EPR valve limits how cold a warmer evaporator can get by maintaining minimum suction pressure above the lower-temperature coil setpoint. Example: produce case at 35°F uses an EPR; freezer evaporator at -10°F does not.

Q43. Answer: D

A relief discharge goes outside, not into the building. CSA B52 requires the pressure-relief device on a refrigerant pressure vessel such as a liquid receiver to discharge to the outdoors, terminating clear of doors, windows and fresh-air intakes so a release cannot re-enter the building or an occupied space. The vent line must be sized and routed so it cannot throttle the device's rated relieving capacity, and it must not be trapped, capped or valved off. Venting into the machinery room is wrong: machinery-room vapour detection and mechanical ventilation are separate requirements that deal with leaks, and they are never a substitute for a relief vent. Piping the relief into the suction line gives no protection at all, because it discharges into the same system it is supposed to protect. Leaving the outlet bare dumps the charge into the plant room. Confirm the termination heights and clearances in the edition of B52 adopted by your province.

Q44. Answer: C

Check cause before replacing compressor. Refrigerant flood-back causes liquid slugging that hydraulically locks compressor pistons. Low voltage at the compressor terminals causes high current + thermal overload. Compressor may free itself after cooling — but replacing without fixing the cause means the replacement fails too. Check: liquid slugging (suction line frost/wet), power supply and voltage at the terminals, overload conditions.

Q45. Answer: D

Liquid refrigerant expands strongly with temperature and is very nearly incompressible, so a liquid-full cylinder has nothing left to absorb that expansion. Leave a vapour space and a warm afternoon simply walks the pressure up the saturation curve, which is what the cylinder is built for. Fill it solid and the same temperature rise has nowhere to go: pressure climbs hydrostatically, far faster and far higher than any saturation pressure, and the cylinder can rupture. That is why recovery cylinders are filled by weight, checked against the tare weight and the capacity marked on the cylinder itself, and why many are fitted with a float or a shut-off that stops the machine before the cylinder fills. Two related rules come from the same physics: never put a flame or other heat source on a cylinder to speed up a transfer, and never refill a disposable single-use cylinder, which has no overfill protection and was not built to be filled again.

Q46. Answer: C

Superheat is heat added to vapour beyond saturated vapour, so on a blend with glide it is measured from the dew temperature: 32°F - 21.9°F, about 10°F. R-407C is a zeotropic blend: at one pressure it starts boiling at the bubble point and finishes at the dew point, and on this chart the two are 11.5°F apart at 40 psig. The refrigerant is only fully vapour once it reaches the dew temperature, so that is the reference for superheat. Subtracting the bubble value gives about 22°F, overstating superheat by the whole glide; a technician who believes that reading may diagnose a starved coil or low charge and add refrigerant to a system that does not need it. The average value still counts part of the boiling range as superheat, giving about 16°F. The bubble column is the right one for subcooling at the condenser outlet, but the liquid line temperature has nothing to do with suction superheat.

Q47. Answer: D

Evaporator icing: most commonly caused by insufficient airflow across the coil (dirty filter, blocked return, or failed fan). Normal operation: evaporator coil surface stays above 32°F (0°C) as warm air flows across it, continuously melting any frost. Restricted airflow (clogged filter, failed blower, blocked return): coil surface temperature drops below freezing, moisture freezes faster than it can drain. Ice eventually insulates the coil completely, stopping heat transfer. Also can be caused by low refrigerant charge (low suction pressure = low coil temp). An overcharge does the opposite — it raises suction pressure and coil temperature, so it floods the compressor rather than icing the coil. A weak run capacitor does not slow the compressor either: a single-phase induction motor runs at essentially fixed speed, and a weak capacitor shows up as hard starting, high amp draw, or overload trips — and any loss of compressor capacity leaves the coil warmer, not frozen. Start diagnosis with: check filter, check blower operation.

Q48. Answer: C

Condenser split is the condensing saturation temperature minus the dry-bulb temperature of the air entering the condenser. An air-cooled condenser can only push its heat into the air it is given, so the entering air is the reference the condensing temperature has to be judged against. A condensing temperature on its own means nothing until you know whether the unit is sitting in cool morning air or in full afternoon sun. A split much wider than the unit was designed for says the coil cannot move its heat: fouled fins, a failed fan or one turning the wrong way, discharge air recirculating around the unit, blocked airflow, or non-condensables occupying condenser volume. A split narrower than design, on a unit that is cooling poorly, points the other way and suggests there is not enough refrigerant in the coil to load it. Note what the other comparisons actually give you: condensing saturation temperature against liquid line temperature is subcooling, a different measurement answering a different question, and wet-bulb belongs to evaporative equipment rather than a dry air-cooled coil.

Q49. Answer: C

Floating head pressure: condenser fan speed or staging is controlled to allow condensing pressure to drop to the lowest stable level based on ambient conditions — improving system efficiency. Traditional systems maintain high fixed head pressure year-round. Floating head control reduces fans (or varies their speed) to allow the head pressure to drop as ambient drops in winter, reducing compressor work (lower pressure ratio = less energy per unit of refrigeration). The float is not unlimited, though: every floating head strategy is given a floor, because the metering device still has to be left with enough pressure difference across it to feed the evaporator, and refrigerant velocity has to stay high enough to carry oil back. The floor is worked out from the system design data for that refrigerant and evaporator temperature, and below it the same head pressure control described elsewhere in this bank takes over and holds condensing pressure up.

Q50. Answer: C

Hot gas defrost: discharge gas heats evaporator from inside — efficient and fast. During hot gas defrost, solenoid valves redirect superheated compressor discharge gas directly to the evaporator inlet (in some designs the coil is fed in reverse flow). The hot gas condenses inside the evaporator coil, releasing its heat of condensation directly to the frost accumulation. This is faster and more energy-efficient than electric defrost. Common in commercial walk-in coolers and display cases.

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